Synopsis: Real-Time Forest Fire Prediction

1. Introduction & Problem Statement

Forest fires represent a severe ecological and socio-economic hazard across India, causing loss of biodiversity, massive carbon emissions, and environmental degradation. Traditional fire detection relies heavily on delayed polar-orbiting satellite passes or manual ground reporting, resulting in critical response latency. Predicting fire vulnerability before ignition requires combining atmospheric dryness, surface temperatures, and local fuel moisture. This project addresses these challenges by constructing an automated, low-latency pipeline that ingests satellite observation streams from ISRO's INSAT-3DS and fuses them with weather parameters for **near real-time (NRT) active fire detection** and **dynamic fire risk indexing**.

2. System Architecture & Why Apache Kafka is Used

- **Why Apache Kafka?**
    1. **Decoupling Ingestion from Processing:** Processing 3207×3062 satellite rasters takes variable CPU/GPU compute time. Kafka acts as a distributed buffer, ensuring that raw file downloads from MOSDAC never block downstream analytical consumers.
    2. **High Throughput & Backpressure Management:** INSAT-3DS publishes full-disk/sector scans every 30 minutes. Kafka efficiently streams millions of pixel-level event payload references without memory spikes.
    3. **Spatial Partitioning (Uber H3 Indexing):** Using H3 spatial cell indices as Kafka message keys guarantees that satellite rasters and weather station feeds for the same forest reserve land on the **same partition**. This enables stateful, low-latency stream joins without expensive cross-cluster shuffling.

3. Analytical Workflow & Key Contributions

1. **Look-Up Table (LUT) Array Decoding:** Fast vector transformation converting raw 10-bit count rasters into physical Brightness Temperature (Kelvin).
2. **Contextual Thermal Detection:** Evaluates radiance divergence between Mid-Wave Infrared (3.9 micrometers) and Thermal Infrared (TIR1 - 10.8 micrometers): $\Delta T = T_{MIR} - T_{TIR1} > 10K$.

4. Academic & Institutional References

1. **Giglio, L., Descloitres, J., Justice, C. O., & Kaufman, Y. J. (2003).** An Enhanced Contextual Fire Detection Algorithm for MODIS. *Remote Sensing of Environment*, 87(2-3), 273–282.
2. **Space Applications Centre (SAC), ISRO (2023).** *INSAT-3DS Meteorological & Oceanographic Satellite Payload Specifications and Data Products Manual*. Meteorological and Oceanographic Satellite Data Archival Centre (MOSDAC).
3. **Kaufman, Y. J., et al. (1998).** Potential Global Fire Monitoring from EOS-MODIS. *Journal of Geophysical Research: Atmospheres*, 103(D24), 32215–32238.

Dataset Description

Primary Dataset: ISRO INSAT-3DS Imager Level-1C Standard Geo-Projected (3SIMG_L1C_SGP)

1. Dataset Overview

The 3SIMG_L1C_SGP dataset is produced by the Space Applications Centre (SAC), ISRO, and archived at MOSDAC. Captured by the geostationary INSAT-3DS satellite located at 82.5 longitude, it provides continuous multispectral coverage over India and the surrounding Asian sector at half-hourly intervals.

2. Spatial and Temporal Specifications

- **Temporal Resolution:** 30 Minutes (48 acquisition scenes per day).
- **Spatial Coverage:** Geo-projected Mercator raster covering longitudes 30 to 130 degrees and latitudes 50 degrees.
- **Raster Dimensions:** 3207×3062 pixels (≈ 9.8 million pixels per scene).
- **Spatial Resolution:** $1km$ for Visible (VIS) channel; $4km$ for Mid-Wave Infrared (MIR) and Thermal Infrared (TIR) channels.
- **File Format:** Hierarchical Data Format 5 (.h5) containing 2D/3D integer rasters and 1D calibration Look-Up Tables (LUT).

3. Key Dataset Channels & Physical Properties

HDF5 Dataset Key	Physical Channel	Waveband Range	Storage Type & Shape	Applied Usage in Project
IMG_MIR	Mid-Wave Infrared (MIR)	3.8-400 micrometers	uint16 raster (1, 3207, 3062)	Primary active fire detector; sensitive to sub-pixel high-temperature combustion.
IMG_MIR_TEMP	MIR Temperature LUT	Physical Kelvin (K)	float32 1D array (1024,)	Look-up array mapping MIR raw counts ($0 -- 1023$)

				directly to Kelvin.
IMG_TIR1	Thermal Infrared 1 (TIR-1)	10.3 - 11.3 micrometers	uint16 raster (1, 3207, 3062)	Measures background surface and cloud top skin temperatures.
IMG_TIR1_TEMP	TIR-1 Temperature LUT	Physical Kelvin (K)	float32 1D array (1024,)	Look-up array mapping TIR-1 counts to Kelvin for background baseline comparison.
IMG_VIS / _ALBEDO	Visible Spectrum (VIS)	0.55 - 0.75 micrometers	uint16 raster / Albedo	Optical imagery used for daytime smoke plume detection and cloud verification.
IMG_SWIR / _ALBEDO	Short-Wave Infrared	$1.55 - 1.70\,\mu\text{m}$	uint16 raster / Albedo	Quantifies vegetation leaf water content and dry fuel dryness.

Sun_Elevation	Solar Elevation Angle	Degrees	float32 raster (1, 3207, 3062)	Applied to calculate solar zenith angles (θ_z) to filter out daytime solar glare.
X, Y, proj_dim	Projection Metadata	Spatial Meters / Extent	1D coordinate vectors	Georeferencing pixel indices (i, j) into latitude and longitude coordinates.

Data Preprocessing & Radiometric Calibration Chain

1. **Raster Reshaping:** Squeezing the singleton batch dimension (1, 3207, 3062) to obtain standard 2D rasters (3207, 3062).
2. LUT Indexing: Transforming integer digital counts (DN) into physical Brightness Temperatures (T_B in Kelvin) via vector indexing:

3. $T_{MIR}(i,j) = LUT_{MIR}\left[clip(DN_{MIR}(i,j), 0, 1023)\right]$
4. Space & Invalid Masking: Filtering space background, missing telemetry frames, and corrupted pixels where $T_B < 180K$ or $T_B > 380K$.
5. Thermal Delta Array Generation: Computing $\Delta T(i,j) = T_{MIR}(i,j) - T_{TIR1}(i,j)$ across the spatial grid to isolate candidate combustion sources.

2. Ancillary & Verification Datasets

A. MOSDAC / IMD Automatic Weather Station (AWS) In-Situ Feeds

- **Source:** ISRO / India Meteorological Department (IMD) Ground Network (mosdac.gov.in/aws).
- **Update Frequency:** Hourly to sub-hourly telemetry.
- **Parameters Captured:** Surface Ambient Temperature ($^\circ\text{C}$), Relative Humidity ($\%$), Wind Speed (m/s), Wind Direction, and Volumetric Soil Moisture (m^3/m^3).

- **Role in System:** Provides continuous atmospheric desiccation features fed into the Kafka raw.weather stream to compute the **Fuel Moisture Index (FMI)** and **Canadian Fire Weather Index (FWI)**.

B. ISRO VEDAS / Bhuvan High-Resolution Vegetation & Fuel Datasets

- **Source:** NRSC Bhuvan & ISRO VEDAS portals (derived from **Resourcesat-2A LISS-3 / AWiFS**).
- **Spatial Resolution:** $23.5mto56m$.
- **Parameters Captured: Normalized Difference Vegetation Index (NDVI)** and **Normalized Difference Water Index (NDWI)**.
- **Role in System:** Serves as a static/weekly baseline layer specifying live fuel load density and canopy moisture stress over Indian forest compartments.

C. NASA FIRMS / VIIRS ($375m$) Active Fire Verification Stream

- **Source:** NASA LANCE / VIIRS (Visible Infrared Imaging Radiometer Suite on S-NPP / NOAA-20).
- **Spatial Resolution:** $375m$ thermal pixel resolution (2–4 passes/day over India).
- **Format:** Near Real-Time REST API returning JSON / GeoJSON point features.
- **Role in System:** Used as high-resolution ground truth to validate broad $4km$ INSAT-3DS detections and train the contextual machine learning models.

Environment Setup and Tool Configuration

1. System Requirements & Foundation Architecture

To process high-dimensional satellite rasters (3207×3062 pixels per channel every 30 minutes) and execute real-time streaming inference, the environment is configured across a hybrid CPU/GPU architecture:

- **Operating System:** Ubuntu 22.04.3 LTS (Jammy Jellyfish) 64-bit / Linux Kernel 5.15+
- **Compute / Memory:** 32-core CPU, 64 GB System RAM, NVIDIA RTX 3090 / A4000 GPU (24 GB VRAM)
- **CUDA / cuDNN Ecosystem:** CUDA Toolkit 11.8, cuDNN v8.6, NVIDIA Driver v525.105.17
- **Virtualization & Package Manager:** micromamba / conda (Python 3.9 runtime isolation)

Bash

```
# Update System Packages & Install Core Development Headers

sudo apt-get update && sudo apt-get install -y \

    build-essential cmake git curl wget unzip \

    libhdf5-dev libnetcdf-dev libgdal-dev gdal-bin \

    openjdk-11-jdk graphviz
```

2. Distributed Event Streaming Pipeline Setup (Apache Kafka)

Apache Kafka decouples heavy satellite file downloading from low-latency streaming consumers.

Step 2.1: Apache Kafka Installation & Service Systemd Configuration

Bash

```
# Download and Extract Apache Kafka 3.6.1 (Scala 2.13)

cd /opt

sudo wget https://archive.apache.org/dist/kafka/3.6.1/kafka_2.13-3.6.1.tgz

sudo tar -xzf kafka_2.13-3.6.1.tgz

sudo mv kafka_2.13-3.6.1 kafka
```

Step 2.2: Kafka Topic Creation & Spatial Partitioning Strategy

Bash

```
# Start Zookeeper and Kafka Broker

/opt/kafka/bin/zookeeper-server-start.sh -daemon /opt/kafka/config/zookeeper.properties

/opt/kafka/bin/kafka-server-start.sh -daemon /opt/kafka/config/server.properties


# Create Production Event Topics with 6 Partitions for Parallel Worker Processing

/opt/kafka/bin/kafka-topics.sh --create --bootstrap-server localhost:9092 \

    --replication-factor 1 --partitions 6 --topic raw.insat.l1c


/opt/kafka/bin/kafka-topics.sh --create --bootstrap-server localhost:9092 \

    --replication-factor 1 --partitions 6 --topic raw.weather.aws


/opt/kafka/bin/kafka-topics.sh --create --bootstrap-server localhost:9092 \

    --replication-factor 1 --partitions 6 --topic fire.alerts.geojson
```

3. Geospatial & Satellite Processing Stack

Satellite data handling requires specific C-compiled libraries (GDAL, HDF5, NetCDF) bound to Python.

Step 3.1: Python Virtual Environment Initialization

Bash

```
# Create and activate dedicated environment via Micromamba

micromamba create -n forest_fire_env python=3.9 -c conda-forge -y

micromamba activate forest_fire_env
```

Install C-bound Geospatial Drivers via Conda-Forge

micromamba install -c conda-forge gdal=3.6.4 h5py=3.8.0 netcdf4=1.6.3 rasterio=1.3.8 pyproj=3.5.0 -y

Step 3.2: Python Data Science & Streaming Stack

Bash

Install Kafka, ML frameworks, and Visualization

pip install confluent-kafka==2.3.0 \

torch==2.0.1+cu118 torchvision==0.15.2+cu118 --index-url https://download.pytorch.org/whl/cu118 \

lightgbm==4.1.0 xgboost==2.0.0 scikit-learn==1.3.0 \

pandas==2.0.3 numpy==1.24.3 matplotlib==3.7.2 seaborn==0.12.2 \

h3==3.7.6 geojson==3.0.1 tqdm==4.65.0

4. Environment Verification Script

Validate that CUDA, HDF5 reading, GDAL, and Kafka connections work properly before running pipelines:

Python

test_env.py

import confluent_kafka

import gdal

import h5py

import h3

import torch

```
print("=== Environment Verification Check ===")
print(f"[x] PyTorch Version: {torch.__version__} | CUDA Available: {torch.cuda.is_available()}")
if torch.cuda.is_available():
    print(f"    GPU Device: {torch.cuda.get_device_name(0)}")
print(f"[x] HDF5 Version: {h5py.__version__}")
print(f"[x] GDAL Version: {gdal.__version__}")
print(f"[x] Uber H3 Spatial Indexing Version: {h3.__version__}")
print(f"[x] Confluent Kafka Client Version: {confluent_kafka.__version__}")
print("=== Environment Check Passed Successfully! ===")
```

Literature Survey

Paper Title	Publication Authors & Journal	Contents & Relevance	Limitations
[1] Development of gridded surface meteorological data for ecological applications and modelling	Abatzoglou, J. T. (2013). *Int. J. of Climatology*, 33, 121-131.	Develops high-resolution gridded meteorological datasets (METDATA) coupling spatial interpolation with reanalysis feeds to drive surface fire weather indices.	Lacks real-time sub-hourly latency required for active fire event dispatches.
[2] A review on forest fire detection techniques	Alkhatib, A. A. A. (2014). *Int. J. of Distributed Sensor Networks*, 10(3).	Surveys terrestrial WSNs, optical satellite imaging, and thermal sensors for wildfire detection.	Focuses heavily on hardware deployments rather than real-time distributed stream data pipelines.
[3] Wildfire spread forecasting with deep learning	Anastasiou, N., & Kondylatos, S. (2026). *IEEE Access*.	Proposes spatio-temporal deep learning architectures for predicting multi-day wildfire burn perimeters.	High computational overhead; sensitive to unobserved localized wind gusts.
[4] Machine learning and deep learning for wildfire spread prediction: A review	Andrianarivony, H. S., & Akhloufi, M. A. (2024). *Fire*, 7(2).	Evaluates CNNs, LSTMs, and Transformers applied to remote sensing time-series for spread modeling.	Does not cover stream-ingestion backpressure management in distributed environments.
[5] A review on	Barmpoutis, P., Papaioannou, P., et	Reviews visible and infrared optical	Vulnerable to false positives induced by

early forest fire detection systems using optical remote sensing	al. (2020). *Sensors*, 20(22), 6442.	imaging techniques for terrestrial and aerial early fire detection.	cloud cover and solar glint reflections.
[6] A systematic review of applications of machine learning techniques for wildfire fire management	Bot, K., & Borges, J. G. (2022). *Inventions*, 7(3), 84.	Categorizes ML applications across fire prevention, real-time detection, and post-fire recovery phases.	Broad conceptual focus without granular technical benchmarks on specific satellite channels.
[7] Artificial neural network-based forest fire detection system using wireless sensor networks	Chauhan, A., Semwal, S., et al. (2013). *IEEE INDICON*.	Implements ANN classifiers on embedded sensor nodes monitoring humidity, temperature, and gas concentrations.	Limited spatial range; physical sensors are easily destroyed by direct flame contact.
[8] A data mining approach to predict forest fires using meteorological data	Cortez, P., & Morais, A. (2007). *EPIA*, 512-523.	Uses Support Vector Machines (SVM) and weather station metrics to predict burned forest area.	Relies on small spatial datasets (Montesinho park) without satellite thermal band integration.
[9] Desert/forest fire detection using machine/deep learning techniques	Davis, M., Shekaramiz, M., et al. (2023). *Fire*, 6(3), 113.	Compares classical ML vs deep CNN vision models across arid desert and dense forest terrain.	High false-alarm rate in desert environments due to sun-baked rock surface heating.
[10] Next day wildfire spread: A machine learning dataset to predict wildfire spreading	Huot, F., Hu, R. L., Lakshmanan, V., & Bhatia, A. (2022). *IEEE TGRS*, 60.	Benchmark dataset combining MODIS/VIIRS thermal anomalies with elevation and weather grids for	Fixed 24-hour prediction step is too coarse for immediate sub-hourly tactical response.

from remote-sensing data		24h spread prediction.	
[11] A survey of distributed data stream processing frameworks	Isah, H., Abughofa, T., Mahfuz, S., et al. (2019). *IEEE Access*, 7, 154300-154316.	Evaluates distributed streaming engines (Kafka, Flink, Storm, Spark Streaming) for low-latency ingest.	General computer science review; lacks domain application to geospatial/raster array streaming.
[12] A review of machine learning applications in wildfire science and management	Jain, P., Coogan, S. C. P., et al. (2020). *Environmental Reviews*, 28(4), 478-505.	Comprehensive review detailing ML models used for fire ignition, fire weather, and spread modeling.	Highlights a persistent literature gap regarding live streaming pipeline integration.
[13] Assessment of deep learning models integrated with weather and environmental variables for wildfire spread prediction	Kim, J., & Hu, Y. (2025). *Natural Hazards*.	Fuses multi-channel remote sensing with ERA5 reanalysis weather data into deep Convolutional networks.	Heavy dependence on high-latency reanalysis datasets restricts real-time operations.
[14] Adversarial concept drift detection under poisoning attacks for robust data streams	Korycki, L., & Krawczyk, B. (2022). *Machine Learning*, 111.	Proposes robust drift detection algorithms for streaming classifiers subject to data corruption and shifts.	Evaluated primarily on synthetic tabular streams rather than satellite imagery feeds.
[15] Combining active learning with concept drift detection for data stream mining	Krawczyk, B., Pfahringer, B., et al. (2018). *IEEE ICMLA*.	Framework for updating streaming classifiers by requesting minimal ground-truth labels during drift events.	Ground-truth fire labels (e.g., ground reports) suffer from significant verification delays.

[16] Kafka: A distributed messaging system for log processing	Kreps, J., Narkhede, N., & Rao, J. (2011). *Proc. NetDB'11*.	Foundational paper introducing Apache Kafka's distributed commit log architecture, pub-sub model, and partitioning.	Explores generic text/log data rather than binary geospatial raster payload arrays.
[17] CNN-BiLSTM: A novel deep learning model for near-real-time daily wildfire spread prediction	Marjani, M., Mahdianpari, M., & Mohammadimanesh, F. (2024). *Remote Sensing*, 16(8), 1357.	Fuses CNN spatial feature extraction with BiLSTM temporal sequence modeling for daily spread prediction.	Operates on daily temporal composites, missing sub-hourly geostationary fire surges.
[18] Daily fire danger prediction in Catalonia using machine learning	Martin-Gomez, V., et al. (2026). *In Press*.	Evaluates localized Random Forest and XGBoost models for predicting daily Fire Danger Indices.	Regional focus restricted to Mediterranean fuel types; requires retraining for tropical forests.
[19] Forest fire monitoring and detection of faulty nodes using wireless sensor networks	Mohapatra, S., & Khilar, P. M. (2016). *IEEE TENCON*.	Applies fault-tolerant clustering in WSNs to distinguish actual thermal fires from sensor hardware errors.	Limited spatial deployment scale compared to country-wide satellite coverage.
[20] Edge computing-based real-time forest fire detection using UAV thermal and color imagery	Mu, L., & Yang, Y. (2025). *IEEE JSTARS*.	Deploys lightweight vision models on drone edge devices (NVIDIA Jetson) for immediate thermal fire spotting.	Constrained flight duration and battery range prevent continuous state-wide monitoring.
[21] Evaluation of distributed stream processing frameworks for IoT applications in	Nasiri, H., Nasehi, S., & Divsalar, M. (2019). *Journal of Big Data*, 6, 61.	Benchmarks throughput, latency, and resource utilization across Kafka, Storm, and	Benchmarked on short structured IoT messages rather than multi-megabyte

Smart Cities		Spark for streaming IoT data.	HDF5 satellite chunks.
[22] Optimizing stacked ensemble machine learning models for accurate wildfire spread prediction	Nguyen Van, L., & Lee, G. (2025). *Remote Sensing*, 17(9).	Combines XGBoost, LightGBM, and CatBoost into a stacked ensemble for predicting fire spread perimeters.	High inference latency makes real-time integration into streaming loops challenging.
[23] FireCast: Leveraging deep learning to predict wildfire spread	Radke, D., Hessler, A., & Long, D. (2019). *IJCAI 2019*, 4575-4581.	Uses fully convolutional networks (FCN) incorporating elevation, wind vectors, and satellite fire historical masks.	Requires dense local GIS feature inputs that are often missing in developing tropical regions.
[24] Artificial intelligence for forest fire prediction	Sakr, G. E., Elhajj, I. H., Mitri, G., & Wehbe, C. S. (2010). *IEEE/ASME AIM*.	Early application of SVMs with weather parameters (temperature, RH, wind) to forecast fire risk zones.	Lacks satellite thermal radiometer feeds, leading to higher false-alarm rates during dry non-fire days.
[25] Predictive modeling of wildfires: A new dataset and machine learning approach	Sayad, Y. O., Mousannif, H., & Al Moatassime, H. (2019). *Fire Safety Journal*, 104, 130-146.	Fuses satellite remote sensing (MODIS) with meteorological data for big data fire risk prediction.	Static batch processing architecture lacks dynamic real-time streaming capability.
[26] Convolutional LSTM network: A machine learning approach for precipitation nowcasting	Shi, X., Chen, Z., Wang, H., et al. (2015). *NeurIPS 28*.	Introduces ConvLSTM to simultaneously model spatial structures and temporal sequences in raster streams.	Computationally intensive; prone to vanishing gradients over extended sequence windows.
[27] Global wildfire	Shmuel, A., &	Generates global	Provides static

susceptibility mapping based on machine learning models	Heifetz, E. (2022). *Forests*, 13(6), 908.	baseline susceptibility maps using Random Forest trained on multi-decadal historical fire records.	long-term risk mapping without sub-hourly dynamic weather adaptation.
[28] Improved lightweight YOLOv11 algorithm for real-time forest fire detection	Tao, Y., & Li, B. (2025). *Electronics*, 14.	Modifies YOLOv11 for lightweight object detection of flames and smoke plumes from optical camera streams.	Optical vision fails under heavy cloud obstruction or thick atmospheric haze.
[29] A performance analysis of fault recovery in stream processing frameworks	van Dongen, G., & Van Den Poel, D. (2021). *IEEE Access*, 9.	Benchmarks state recovery times and fault tolerance of stream processing systems during worker node failures.	Focuses on infrastructure recovery without addressing domain-specific geospatial data corruption.
[30] DPMNet: A remote sensing forest fire real-time detection network driven by dual-perspective multiscale features	Wang, G., & Li, H. (2025). *IEEE TCAS-I*.	Proposes DPMNet to extract multi-scale spatial features for identifying small sub-pixel active fire points.	Requires high-resolution optical input, making it less effective on coarse geostationary rasters.
[31] Data stream mining: methods and challenges for handling concept drift	Wares, S., Isaacs, J., & Brooks, E. (2019). *SN Applied Sciences*, 1, 242.	Reviews adaptive algorithms for maintaining ML accuracy when streaming data distributions shift over time.	Theoretical review; does not provide direct implementations for satellite radiance calibration shifts.
[32] Discretized streams:	Zaharia, M., Das, T., Li, H., et al. (2012).	Introduces Discretized Streams	Micro-batching adds artificial latency

Fault-tolerant streaming computation at scale	*NSDI'12*.	(D-Streams) in Spark Streaming for micro-batch stream fault tolerance.	compared to Kafka's native record-at-a-time streaming.
[33] An enhanced wildfire spread prediction using multimodal satellite imagery and deep learning	Zakari, R., & Malik, O. A. (2025). *Remote Sens. Appl.: Soc. Environ.*.	Integrates multi-modal optical, thermal, and SAR satellite imagery into a deep network for spread mapping.	SAR and optical alignment introduces high data pre-processing latency.